

Solving Quadratic Equations

Algebra Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Solve quadratic equations by isolating x squared and taking the square root of both sides
- Identify when a quadratic equation yields real, imaginary, or repeated solutions
- Solve quadratic equations in factored form and vertex form

Problems

1. Solve for x:

$$x^2 + 9 = 9$$

2. Solve for x:

$$x^2 - 7 = 29$$

3. Solve for x:

$$x^2 + 12 = 61$$

4. Solve for x:

$$4x^2 = 100$$

5. Solve for x:

$$49x^2 = 1$$

6. Solve for x. Identify whether the solutions are real or imaginary:

$$9x^2 + 5 = 2$$



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7. Solve for x:

$$(x - 5)^2 = 16$$

8. Solve for x:

$$(x + 4)^2 = 7$$

9. Solve for x by factoring:

$$x^2 + 3x - 18 = 0$$

10. Solve for x by factoring:

$$2x^2 - x - 15 = 0$$



Solving Quadratic Equations — Answer Key

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Answer Key

1. Answer: $x = 0$

- Subtract 9 from both sides: $x^2 = 0$
- Take the square root of both sides: $x = 0$ (zero is neither positive nor negative, so there is only one solution)

2. Answer: $x = 6$ or $x = -6$

- Add 7 to both sides: $x^2 = 36$
- Take the square root of both sides: $x = \pm 6$

3. Answer: $x = 7$ or $x = -7$

- Subtract 12 from both sides: $x^2 = 49$
- Take the square root of both sides: $x = \pm 7$

4. Answer: $x = 5$ or $x = -5$

- Divide both sides by 4: $x^2 = 25$
- Take the square root of both sides: $x = \pm 5$

5. Answer: $x = 1/7$ or $x = -1/7$

- Divide both sides by 49: $x^2 = 1/49$
- Take the square root of both sides: $x = \pm\sqrt{1/49} = \pm 1/7$

6. Answer: $x = \pm i\sqrt{3}/3$ (imaginary solutions)

- Subtract 5 from both sides: $9x^2 = -3$
- Divide both sides by 9: $x^2 = -1/3$
- Take the square root of both sides: $x = \pm\sqrt{-1/3} = \pm i/\sqrt{3} = \pm i\sqrt{3}/3$ (imaginary)

7. Answer: $x = 9$ or $x = 1$

- Take the square root of both sides: $x - 5 = \pm 4$
- Add 5 to both sides: $x = 5 + 4 = 9$ or $x = 5 - 4 = 1$

8. Answer: $x = -4 + \sqrt{7}$ or $x = -4 - \sqrt{7}$

- Take the square root of both sides: $x + 4 = \pm\sqrt{7}$
- Subtract 4 from both sides: $x = -4 \pm \sqrt{7}$

9. Answer: $x = 3$ or $x = -6$

- Find two numbers that multiply to -18 and add to 3: those are 6 and -3
- Factor: $(x + 6)(x - 3) = 0$
- Set each factor equal to zero: $x + 6 = 0 \rightarrow x = -6$ or $x - 3 = 0 \rightarrow x = 3$

10. Answer: $x = 3$ or $x = -5/2$



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- Find two numbers that multiply to -30 and add to -1: those are -6 and 5
 - Rewrite the middle term and factor by grouping: $2x^2 - 6x + 5x - 15 = 2x(x - 3) + 5(x - 3) = (2x + 5)(x - 3)$
 - Set each factor equal to zero: $x = 3$ or $x = -5/2$
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